

SAT UZBEKISTAN

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1.

A total of 50 people attended a conference and were offered 4 types of entrees. The table shows the number of people who chose each type of entree.

Type of entree	Number of people
Beef	20
Chicken	14
Fish	7
Vegetarian	9
Total	50

If one of these people is selected at random, what is the probability of selecting a person who chose a vegetarian entree?

(A) $\frac{9}{100}$

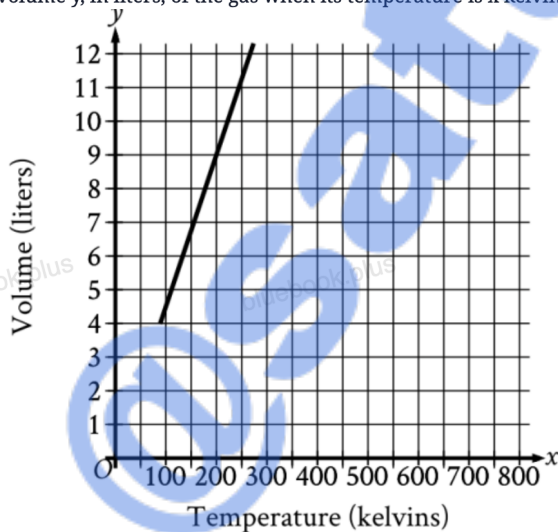
(B) $\frac{9}{50}$

(C) $\frac{1}{4}$

(D) $\frac{9}{10}$

2.

A certain gas is placed inside a container and kept at a constant pressure. The graph shows the estimated volume y , in liters, of the gas when its temperature is x kelvins.



(A) $\frac{200}{9}$

(B) 9

(C) $\frac{9}{200}$

(D) 0

What is the estimated volume, in liters, of the gas when its temperature is 200 kelvins?

3.

During a trip, Bill drove his car for 22.0 miles and then began driving at a constant speed of 70.0 miles per hour. After 2 hours of driving at this speed, what is the total distance, in miles, Bill has traveled during this trip?

4.

x	y
0	17
1	24
2	31

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . What is the y -coordinate of the y -intercept of the line in the xy -plane that represents this relationship?

5.

The function $f(x) = 180(x - 2)$ gives the sum of the interior angles, in degrees, for a polygon with x sides. What is the sum of the interior angles, in degrees, for a polygon with 34 sides?

6.

$$y = 7,700(0.86)^x$$

The given equation estimates the value, in dollars, of a certain car x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the car at the time it was purchased?

(A) 770

(B) 6,622

(C) 7,700

(D) 8,600

7.

At how many points do the graphs of the equations $y = x + 23$ and $y = 12x$ intersect in the xy -plane?

☐ A 0

☐ B 1

☐ C 2

☐ D 12

8.

A space probe uses a square-shaped solar sail to move through space. The side length of the solar sail is $8.75w$ meters, where w is the width, in meters, of the space probe. Which equation gives the area A , in square meters, of the solar sail?

☐ A $A = (w + 8.75)(w + 8.75)$

☐ B $A = (4)(8.75w)$

☐ C $A = (8.75w)(8.75w)$

☐ D $A = \left(\frac{8.75}{w} \right) \left(\frac{8.75}{w} \right)$

9.

The length of a rectangle is 55 inches and the width is x inches. The perimeter is at most 150 inches. Which inequality represents this situation?

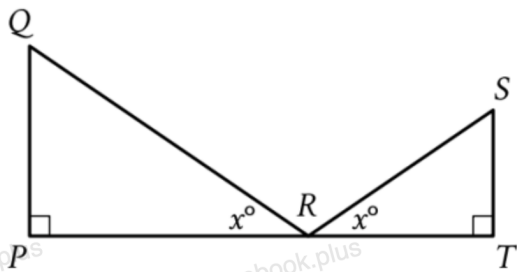
☐ A $2x + 110 \leq 150$

☐ B $2x + 110 \geq 150$

☐ C $2x + 55 \leq 150$

☐ D $2x + 55 \geq 150$

10.



$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\overline{ST}$, $\overline{QP}$, $\overline{PR}$, and $\overline{QR}$ in the figure are 11, 18, 24, and 30, respectively. What is the length of $\overline{SR}$?

(A) $\frac{330}{18}$

(B) $\frac{330}{24}$

(C) $\frac{198}{24}$

(D) $\frac{198}{30}$

11.

$$f(x) = (x - 66)(x - 68)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

(A) 68

(B) 67

(C) 66

(D) -1

12.

A scientist studying the life cycle of butterflies counted the number of butterflies in a certain habitat each day for 46 days. On June 15, there were 114 butterflies in the habitat. The percent increase in the number of butterflies in the habitat from May 1 to June 15 was 18.75%. How many butterflies were in the habitat on May 1?

(A) 6

(B) 21

(C) 93

(D) 96

13.

A moving truck rental company charges its customers \$195 for the first hour of renting a truck and \$75 for each additional hour, plus the cost of gas used. Which equation represents this situation, where y is the total cost, in dollars, of renting a truck from this company for x hours, w is the cost, in dollars, of gas used, and $x \geq 1$?

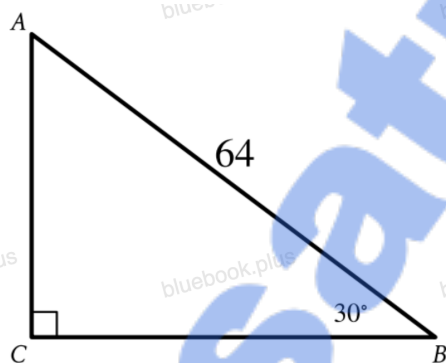
(A) $y = 195 + 75(x - 1) + w$

(B) $y = 195 + 75x + w$

(C) $y = 195 + 75 + w$

(D) $y = 75(x - 1) + w$

14.



Note: Figure not drawn to scale.

Right triangle ABC is shown. What is the value of $\tan A$?

(A) $\frac{\sqrt{3}}{64}$

(B) $\frac{1}{\sqrt{3}}$

(C) $\sqrt{3}$

(D) $32\sqrt{3}$

15.

$f(x)=(x-a)(x-b)$

The function f is defined by the given equation, where a and b are integer constants. If $f(15) > 0$, $f(18) < 0$, and $f(21) > 0$, which of the following could be the value of $a + b$?

- (A) 18
- (B) 20
- (C) 33
- (D) 35

16.

$x^2 + 9x + y^2 - 7y + 11 = 0$

In the xy -plane, the graph of the given equation is a circle. What is the radius of the circle?

- (A) $\frac{\sqrt{86}}{2}$
- (B) $\frac{\sqrt{238}}{2}$
- (C) $\sqrt{11}$
- (D) 11

17.

The table shown summarizes the distribution of 73 employees based on the number of complete years they have worked at a certain company.

Number of complete years at the company	Number of employees
0 to 4	18
5 to 8	4
9 to 12	14
13 to 16	6
17 to 20	16
21 to 24	7
25 to 28	8

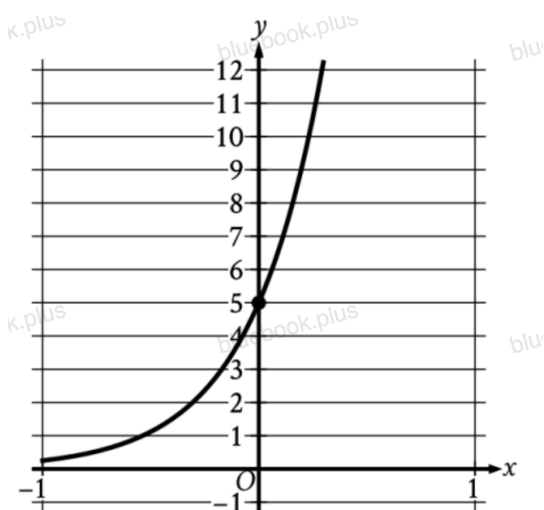
How many of the 73 employees have worked at the company for 17 or more complete years?

18.

The length, y , of a beluga whale was 157 centimeters (cm) when it was born and increased an average of 4.6cm per month for the first 12 months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born?

- ☐ A $y = 157x$
- ☐ B $y = 157x + 157$
- ☐ C $y = 4.6x + 4.6$
- ☐ D $y = 4.6x + 157$

19.



The graph of the exponential function f is shown, where $y = f(x)$. The y -intercept of the graph is $(0, y)$. What is the value of y ?

20.

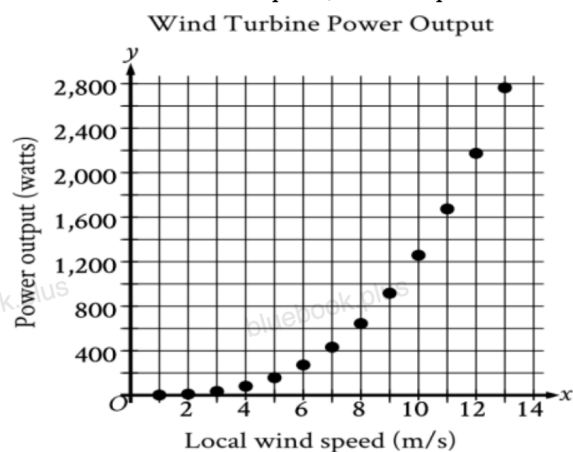
$$g(x) = \frac{1}{11} (5(2)^x + 4)$$

The function g is defined by the given equation. For all values of x , the value of $g(x)$ is greater than k , where k is a constant. What is the greatest possible value of k ?

- ☐ A 0
- ☐ B $\frac{4}{11}$
- ☐ C $\frac{9}{11}$
- ☐ D 4

21.

The scatterplot shows the power output, in watts, of a wind turbine for some local wind speeds, in meters per second (m/s).



Which of the following is closest to the power output, in watts, of the wind turbine when the local wind speed is 8 m/s?

- ☐ A 640
- ☐ B 800
- ☐ C 840
- ☐ D 900

22.

An electric food processor chops 12 pounds of vegetables in 2 minutes. At this rate, how many minutes does it take the processor to chop 72 pounds of vegetables?

23.

For students at a school, the table summarizes the distribution of grade level and location during their lunch break on a certain day.

	Grade 9	Grade 10	Grade 11	Grade 12	Total
On campus	246	231	126	50	653
Off campus	51	55	138	203	447
Total	297	286	264	253	1,100

A student from this school is selected at random. Based on the table, what is the probability of selecting a student that stayed on campus during their lunch break, given that the student is in grade 10 or 11 ? (Express your answer as a decimal or fraction, not as a percent.)

24.

$$x^2 - 97 = -97 + x^2$$

How many distinct real solutions does the given equation have?

- (A) Zero
- (B) Exactly one
- (C) Exactly two
- (D) Infinitely many

25.

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y , in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 2.75 inches, it has an estimated breaking strength of 9,528.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 5.50 inches?

26.

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{394})$. If $h(m) = 0$, what is the greatest possible value of m ?

27.

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

I. $f(x) = a(4.2)^{x+b}$
 II. $g(x) = a(4.2)^x + b$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

28.

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(18) > 0$, $f(21) < 0$, and $f(24) > 0$, which of the following could be the value of $a + b$?

(A) 21

(B) 23

(C) 39

(D) 41

29.

The function f is defined by $f(x) = 9x^2 + kx - 21$, where k is a constant. In the xy -plane, the graph of $y = f(x)$ passes through the points $(r, 23)$ and $(2r, 139)$, where r is a constant. Which of the following could be the value of k ?

(A) 4

(B) 12

(C) 116

(D) 162

30.

The functions f and g are defined by the equations below, where a and b are integer constants, with $a < b$ and $b < 0$. If the equations $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the given equations explicitly shows the y -coordinate of the y -intercept as a constant or coefficient?

1. $f(x) = a(2.7)^{x+b}$

2. $g(x) = a(2.7)^x + b$

A) I only

B) II only

C) I and II

D) Neither I nor II

31.

Tanya drove on the highway and on local roads to complete a trip of 190 miles. The drive took 4 hours. She drove an average speed of 55 miles per hour (mph) on the highway and an average speed of 25 mph on local roads. If x is the time, in hours, Tanya drove on the highway and y is the time, in hours, she drove on local roads, which system of equations represents this situation?

(A) $55x + 25y = 4$
 $x + y = 190$

(B) $55x + 25y = 190$
 $x + y = 4$

(C) $25x + 55y = 4$
 $x + y = 190$

(D) $25x + 55y = 190$
 $x + y = 4$

32.

A circle in the xy -plane has its center at $(14, 13)$ and has a radius of $6k$. Which equation represents this circle?

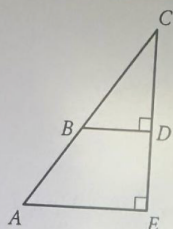
(A) $(x - 14)^2 + (y - 13)^2 = 36k$

(B) $(x - 14)^2 + (y - 13)^2 = 36k^2$

(C) $(x - 14)^2 + (y - 13)^2 = 6k$

(D) $(x - 14)^2 + (y - 13)^2 = 6k^2$

33.



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 25(BD)$. What is the measure of angle CAE ?

(A) $(25 \cdot 57)^\circ$

(B) $(25 + 57)^\circ$

(C) 57°

(D) 25°

34.

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 2)$ and passes through the points $(2, 24)$ and $(-1, 90)$. What is the value of $f(-2) - f(0)$?

(A) 68

(B) 112

(C) 176

(D) 200

35.

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 104(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 104-gram sample starts to decay. Which statement is the best interpretation of $s(58) = 26$ in this context?

(A) Approximately 26 grams of the sample remains 58 years after the sample starts to decay.

(B) The mass of the sample has decreased by approximately 26 grams 58 years after the sample starts to decay.

(C) The mass of the sample has decreased by approximately 58 grams 26 years after the sample starts to decay.

(D) Approximately 58 grams of the sample remains 26 years after the sample starts to decay.

36.

$$g(x) = \frac{1}{5}(5)^{x+4}$$

For the given function g , which of the following equivalent forms shows the y -coordinate of the y -intercept of the graph of $y = g(x)$ in the xy -plane as a constant or coefficient?

A) $g(x) = 125(5)^x$

B) $g(x) = 25(5)^{x+1}$

C) $g(x) = 5(5)^{x+2}$

D) $g(x) = (5)^{x+3}$

37.

$$y = 576^{(2x+2)}$$

The graph of the given equation in the xy -plane has a y -intercept of (r, s) . Which of the following equivalent equations displays the value of s as a constant, a coefficient, or the base?

- A) $y = 331,776^{(x+1)}$
- B) $y = 24^{(4x+4)}$
- C) $y = \frac{1}{24}(24)^{(4x+5)}$
- D) $y = \frac{1}{576}(576)^{(2x+3)}$

38.

A factory makes 9-inch, 7-inch, and 4-inch concrete screws. During a certain day, the number of 9-inch concrete screws that the factory makes is 5 times the number n of 7-inch concrete screws, and the number of 4-inch concrete screws is 22. During this day, the factory makes 100 concrete screws total. Which equation represents this situation?

- A) $9(5n) + 7n + 4(22) = 100$
- B) $9n + 7n + 4n = 100$
- C) $5n + 22 = 100$
- D) $6n + 22 = 100$

39.

A rectangular region of land is divided into 86 square lots of equal area A , in square units. The length of the region is 21.5 times the width of the region. If the length of the region is $x\sqrt{A}$ units, what is the value of x ?

A rectangular region of land is divided into 90 square lots of equal area R , in square units. The length of the region is 2.5 times the width of the region. If the width of the region is $x\sqrt{R}$ units, what is the value of x ?

39-misol ga o'xshash lekin ko'pchilik ikkala masalani ham e'tibor qilmasdan bir xil yechib xatoga yo'l qo'yadi

40.

A grove has 6 rows of birch trees and 5 rows of maple trees. Each row of birch trees has 8 trees 20 feet or taller and 6 trees shorter than 20 feet. Each row of maple trees has 9 trees 20 feet or taller and 7 trees shorter than 20 feet. A tree from one of these rows will be selected at random. What is the probability of selecting a maple tree, given that the tree is 20 feet or taller?

- A. $\frac{9}{164}$
- B. $\frac{3}{10}$
- C. $\frac{15}{31}$
- D. $\frac{9}{17}$

41.

$$y = 2x^2 - 21x + 64$$

$$y = 3x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

- A. -8
- B. -6
- C. 6
- D. 8

42.

Two identical rectangular prisms each have a height of 90 centimeters (cm). The base of each prism is a square, and the surface area of each prism is $K \text{ cm}^2$. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K \text{ cm}^2$. What is the side length, in cm, of each square base?

- A. 4
- B. 8
- C. 9
- D. 16

43.

$$5x + 7y = 1$$

$$ax + by = 1$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

(A) $10x + 7y = 1$
 $ax - 2by = 1$

(B) $10x + 7y = 1$
 $ax + 2by = 1$

(C) $10x + 7y = 1$
 $2ax + by = 1$

(D) $5x - 7y = 1$
 $ax + by = 1$

44.

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water and then reached the surface of the water 12 minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water?